# CHEMISTRY-I

## Theory of Dilute Solutions


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# COLLIGATIVE PROPERTIES

Dilute solutions containing non-volatile solute exhibit the following properties :

(1) Lowering of the Vapour Pressure

(2) Elevation of the Boiling Point

(3) Depression of the Freezing Point

(4) Osmotic Pressure

Colligative Properties (**Greek colligatus** = Collected together)

**Which depends on the number of particles in solution and not in any way on the size or chemical nature of the particles.**

# LOWERING OF VAPOUR PRESSURE : RAOULT'S LAW

The vapour pressure of a pure solvent is decreased when a non-volatile solute is dissolved in it. If **p** is the vapour pressure of the solvent and $\mathbf{p_s}$ that of the solution, the lowering of vapour pressure is $\mathbf{(p - p_s)}$**.**This lowering of vapour pressure relative to the vapour pressure of the pure solvent is termed the Relative lowering of Vapour pressure. Thus,

Relative Lowering of Vapour $\mathbf{(p - p_s)/p}$

As a result of extensive experimentation, Raoult (1886) gave an empirical relation connecting the relative lowering of vapour pressure and the concentration of the solute in solution. This is now referred to as the Raoult's Law

**The relative lowering of the vapour pressure of a dilute solution is equal to the mole fraction of the solute present in dilute solution.**

Raoult's Law can be expressed mathematically in the form :

$$\mathbf{p - p_s / p = n / n + N}$$

Where **n** = number of moles or molecules of solute

**N** = number of moles or molecules of solvent

# Derivation of Raoult's Law

- The vapour pressure of the solution is, therefore, determined by the number of molecules of the solvent present at any time in the surface which is proportional to the mole fraction. That is,

$$\mathbf{p_s \propto N/\ n + N}$$

where N = moles of solvent and n = moles of solute.

$$\mathbf{p_s = \mathit{k} \times N/\ n + N \quad \ldots.\ (1)}$$

*k* being proportionality factor.

In case of pure solvent **n = 0** and hence

Mole fraction of solvent **= N/n + N= N/ 0+N = 1**

Now from equation (1), the vapour pressure **p = k**

Therefore the equation (1) assumes the form

$$p_s = p \times N/\ n + N$$

$$p_s / p = N/\ n + N$$

$$1 - p_s /p = 1 - N/\ n + N$$

$$p - p_s /p = n /\ n + N$$

**This is Raoult's Law**

# Determination of Molecular Mass from Vapour Pressure Lowering

If in a determination w grams of solute is dissolved in W grams of the solvent, m and M are molecular masses of the solute and solvent respectively, we have

No. of Moles of solute (n) = w/m

$$\frac{p - p_s}{P} = \frac{n}{n + N}$$

No. of M ... W/M

Substitu ... Raoult's law Equation

$$\frac{p - p_s}{P} = \frac{w/m}{w/m + W/M}$$

$$\frac{p - p_s}{P} = \frac{wM}{mW}$$

# ELEVATION OF BOILING POINT

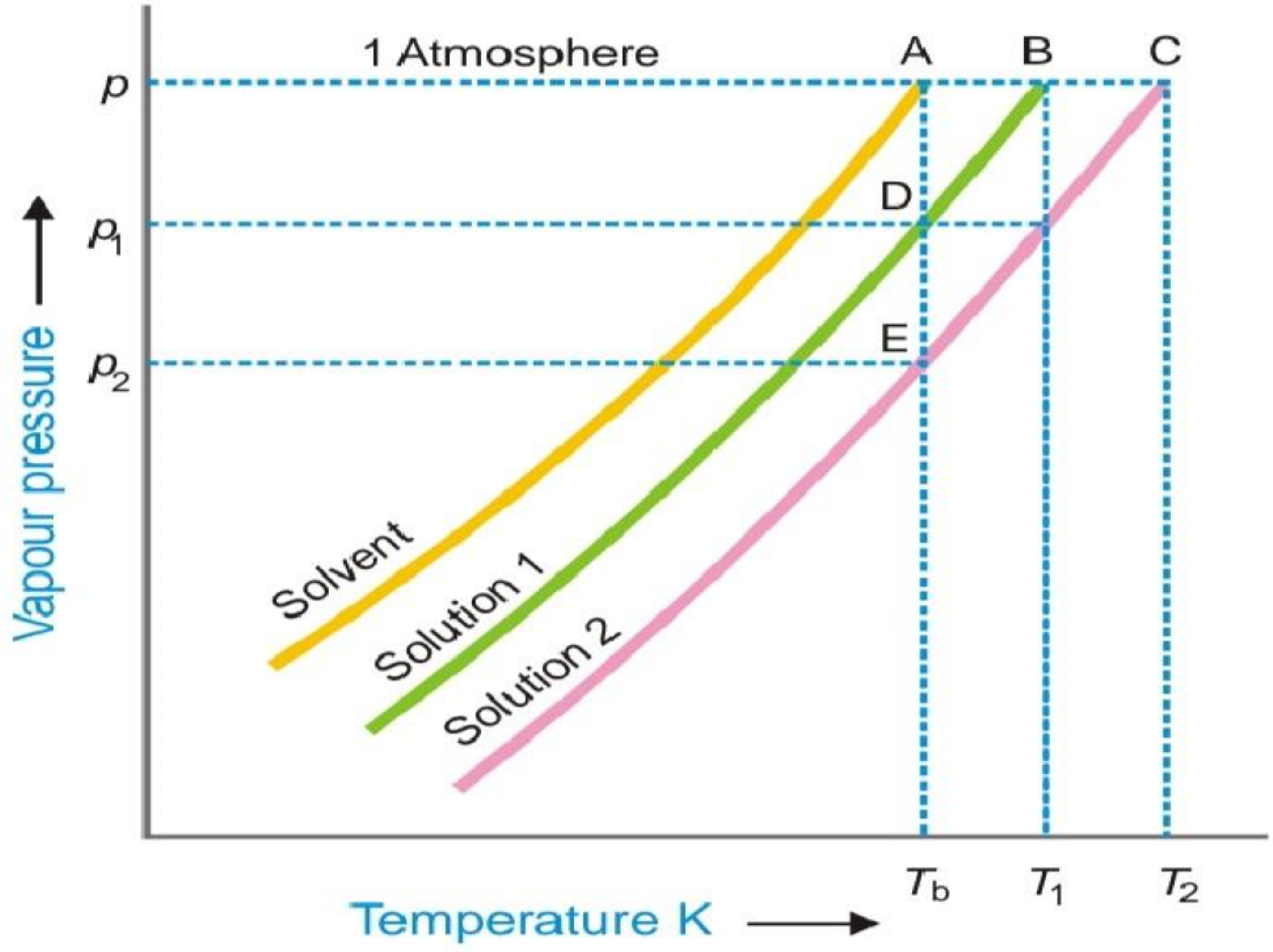
## Relation between Elevation of Boiling Point and Lowering of Vapour pressure

When a liquid is heated, its vapour pressure rises and when it equals the atmospheric pressure, the liquid boils. The addition of a non volatile solute lowers the vapour pressure and consequently elevates the boiling point as the solution has to be heated to a higher temperature to make its vapour pressure become equal to atmospheric pressure. If $T_b$ is the boiling point of the solvent and T is the boiling point of the solution, the difference in the boiling points **($\Delta T$)** is called the elevation of boiling point

$$\mathbf{T - T_b = \Delta T}$$

The vapour pressure curves of the pure solvent, and solutions with different concentration of solute are shown in Fig

# Fig : Elevation of Boiling Point



# Determination of Molecular Mass from Elevation of Boiling Point

The elevation of boiling point is directly proportional to the lowering of vapour pressure.

or $\mathbf{\Delta T \propto p - p_s}$ **...(1)**

Since p is constant for the same solvent at a fixed temperature, from (1) we can write

$\mathbf{\Delta T \propto p - p_s /p}$ **…(2)**

But from Raoult's Law for dilute solutions

$\mathbf{p - p_s / p \propto w\ M/\ W\ m}$ **…(3)**

Since M (mol mass of solvent) is constant from (3)

$\mathbf{p - p_s /p \propto w\ /W\ m}$ **… (4)**

From (2) and (4)

$\mathbf{\Delta T \propto w/m \times 1/\ W}$

$\mathbf{\Delta T = K_b \times w/m \times 1/\ W}$ **… (5)**

where $K_b$ is a constant called **Boiling point constant** or **Ebulioscopic constant of molal elevation constant. If w/m = 1, W = 1, $K_b$ = ΔT.** Thus,

**Molal elevation constant may be defined as the boiling-point elevation produced when 1 mole of solute is dissolved in one kg (1000 g) of the solvent.**

If the mass of the solvent (W) is given in grams, it has to be converted into kilograms. Thus theexpression (5) assumes the form

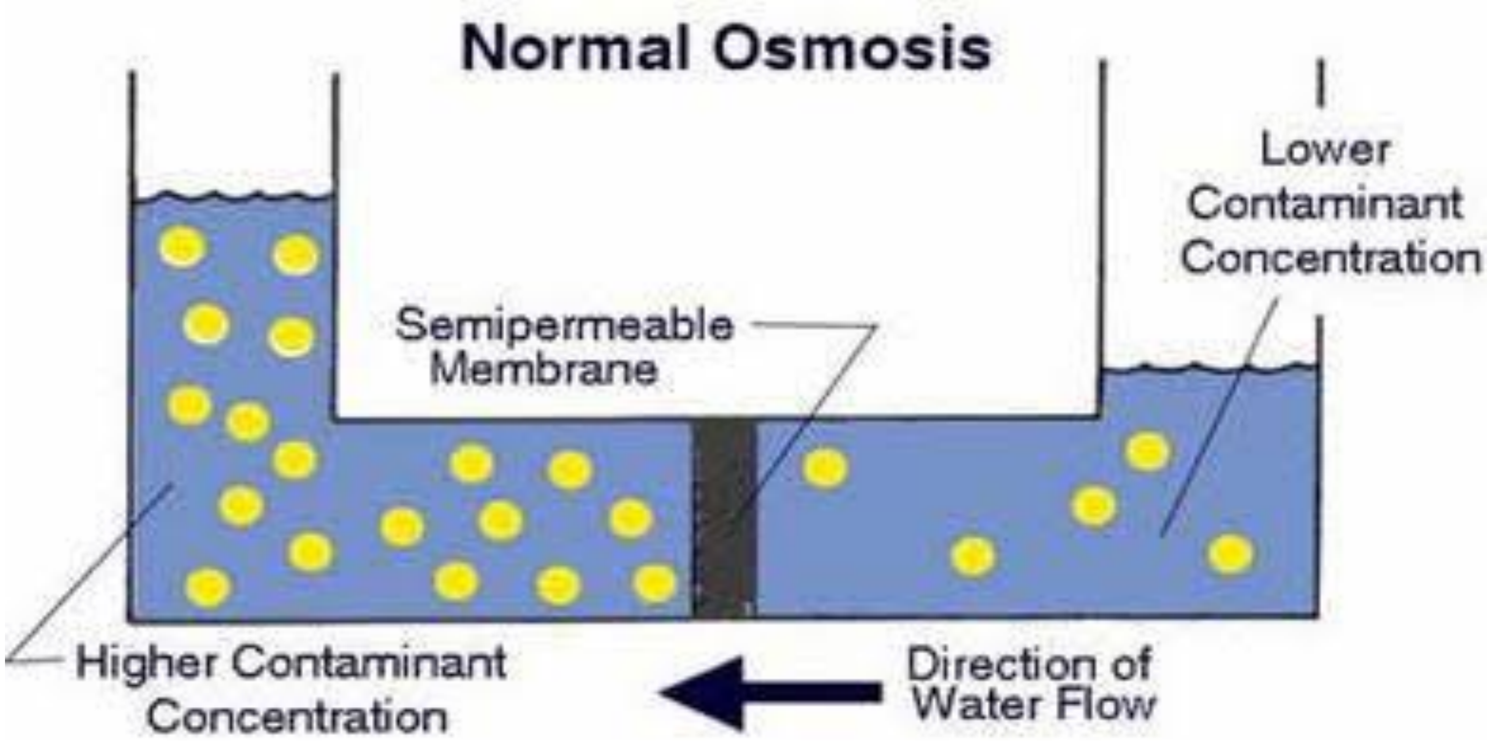
$$\Delta T = K_b \times \frac{w}{m} \times \frac{1}{W/1000}$$

$$m = \frac{1000 \times K_b \times w}{\Delta T \times W}$$

# Osmosis

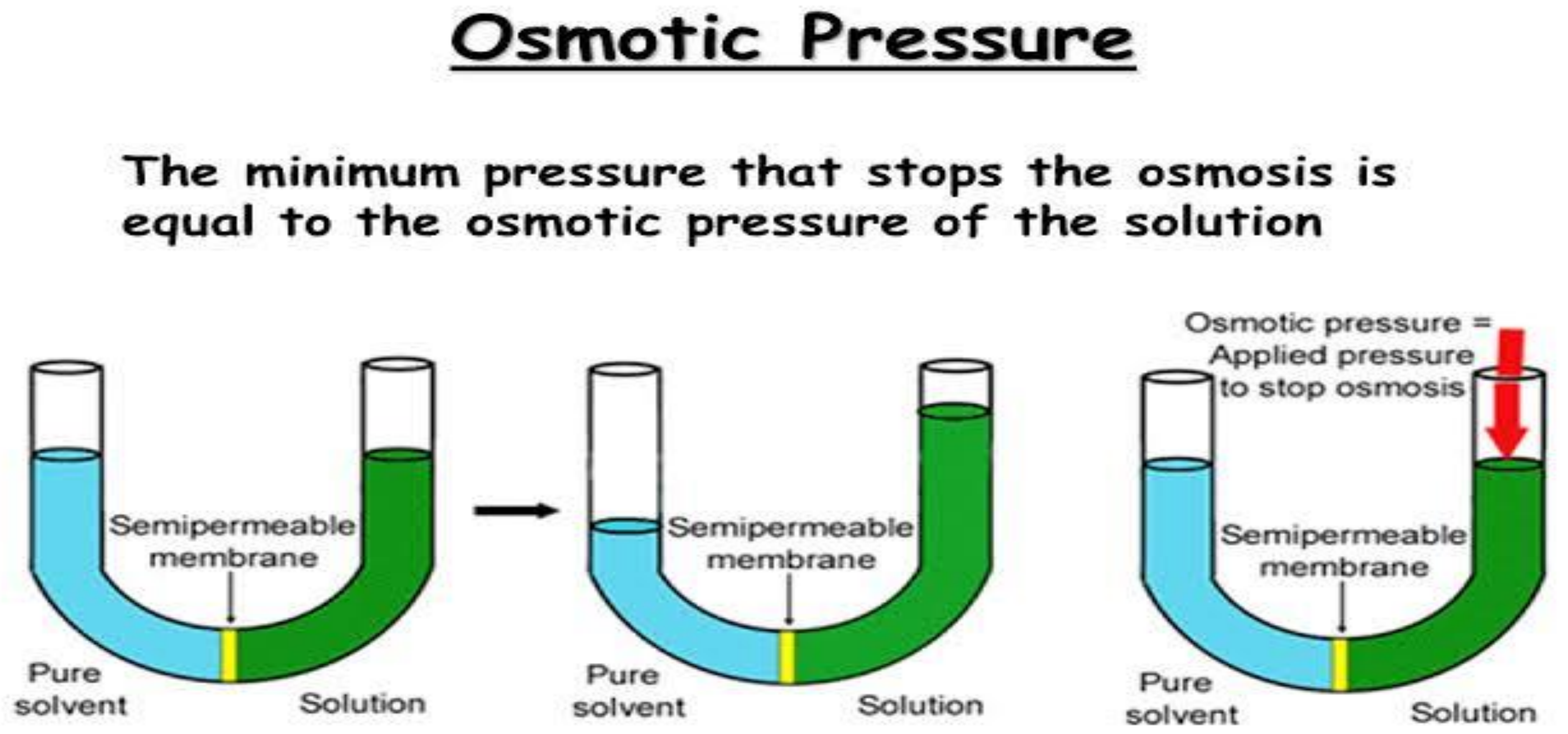
**A membrane which is permeable to solvent and not to solute, is called a semipermeable membrane.**

**The flow of the solvent through a semipermeable membrane from pure solvent to solution, or from a dilute solution to concentrated solution, is termed Osmosis (Greek Osmos = to push).**



# Osmotic Pressure

**The external pressure applied to the solution in order to stop the osmosis of solvent into solution separated by a semipermeable membrane.**



# LAWS OF OSMOTIC PRESSURE

From a study of the experimental results obtained by Pfeffer, van't Hoff showed that for dilute solutions :

**(a) The osmotic pressure of a solution at a given temperature is directly proportional to its concentration.**

**(b) The osmotic pressure of a solution of a given concentration is directly proportional to the absolute temperature.**

From the above findings, van't Hoff (1877) established the laws of osmotic pressure and pointed out that these were closely related to the gas laws.

## (1) Boyle-van’t Hoff Law for Solutions

If $\pi$ is the osmotic pressure and C its concentration, from (a) we can write $\pi \propto C$, if temperature is constant.

If the concentration of the solute is expressed in moles per litre and V is the volume of the solution

that contains 1 mol of solute

$$\mathbf{C = 1 / V}$$

Thus $\boldsymbol{\pi \propto 1 / V}$ **at constant temperature**

This relationship is similar to the Boyle’s law for gases and is known as the Boyle-van’ t Hoff law for solutions.

## (2) Charles-van’t Hoff Law for Solutions

If T is the absolute temperature, from the statement (b), we can write $\boldsymbol{\pi \propto T}$**,** if temperature is constant

This relationship is similar to the Charles’ Law for gases and is known as Charles-van’t Hoff law for solutions.

## (3) Van' t Hoff Equation for Solutions

As shown above the osmotic pressure ($\pi$) of a dilute solution is inversely proportional to the volume (V) containing 1 mole of the solute and is directly proportional to the absolute temperature (T).

This is,

$$\pi \propto 1/V \quad \ldots(1)$$

$$\pi \propto T \quad \ldots(2)$$

Combining (1) and (2) van't Hoff gave the general relationship

$$\pi V = R'T \quad \ldots(3)$$

where R' is a constant. He showed that this equation was parallel to the general Gas Equation **(PV = RT),** as the value of R' calculated from the experimental values of $\pi$, V, and T came out to be almost the same as of the Gas constant, R.It is noteworthy that the van't Hoff Equation (3) was derived for 1 mole of solute dissolved in V litres. If n moles of solute are dissolved in V litres of solution, this equation assumes the form

$$\pi V = nRT$$

**Problem #A solution of cane-sugar (mol mass = 342) containing 34.2 g litre$^{-1}$ has an osmotic pressure of 2.4 atm at 20°C. Calculate the value of R in litre-atmospheres.**

SOLUTION:

From van't Hoff equation

**R=π V/T …(1)**

Where π = osmotic pressure, V = volume of solution in litres containing 1 mole of solute, T = absolute temperature.

In the present case,

Π = 2.4 atm, V = 1/34.2×342 =10 litres,T = 20 + 273 = 293 K

Substituting the values in the expression (1),

R = 0.0819 lit-atm $K^{-1}$ $mol^{-1}$ (Ans.)

# ASSIGNMENT

1.Define or explain the following terms :

(a)Colligative properties (b) Raoult's law (C) Boiling point elevation

2.How is the molecular mass of a solute determined from elevation of boiling point?

3.Boiling point of a liquid increases on adding non-volatile solute in it.

4.18.2 g of urea is dissolved in 100 g of water at 50°C. The lowering of vapour pressure produced is 5 mm Hg. Calculate the molecular mass of urea. The vapour pressure of water at 50°C is 92 mm Hg .(Ans:57.05)

5. Acetone boils at 56.38°C and a solution of 1.41 grams of an organic solid in 20 grams of acetone boils at 56.88°C. If K for acetone per 100 g is 16.7, calculate the mass of one mole of the organic solid.

6.What do you mean by osmosis? State van't Hoff's law of osmotic pressure and deduce osmotic pressure equation p = CRT. The symbols have their usual significance.

7. Calculate the osmotic pressure of a 5% solution of glucose at 18°C.(R = 0.082 atm litre mol–1 deg–1) Ans.6.628 atm